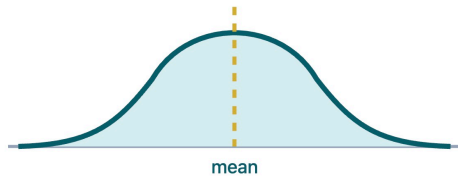


RSR Week 3 : Normal Distributions
and Sampling Distributions

1. A clinic records the resting heart rates of thousands of healthy adults. Which description matches the resulting normal distribution? (referring to image)

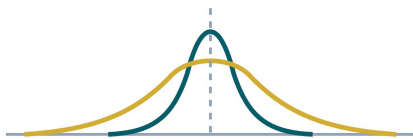


a) Every value is equally common.

b) Symmetric, values cluster around the mean, fewer values far away. ✓

c) Most values sit in the tails with a dip in the middle; asymmetric at the extremes.

2. A delivery company tracks parcel delivery time, with mean $\mu = 40$ mins and SD σ . What happens to the curve if σ gets larger while μ stays the same?



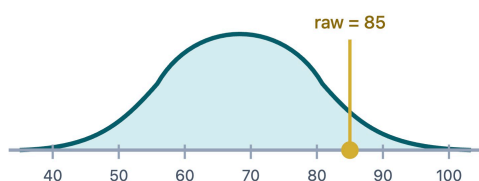
a) The curve slides right.

b) The curve becomes taller and narrower.

c) The curve becomes wider and flatter. ✓

d) The curve slides left.

3. A student scores 85 on a test where the class mean is 70 and the SD is 10. What is the z-score?



a) $z = 0.15$

b) $z = 1.5$ ✓

c) $z = 10$

d) $z = -1.5$

4. Different researchers draw samples of 25 post-operative patients from a hospital. Each sample's average recovery time (5.8 days, 7.1, 6.4, ...). What is the histogram of sample means called?



- a) The data distribution of individual patient recovery times.
- b) The sampling distribution of the sample mean. ✓
- c) The standard normal distribution.

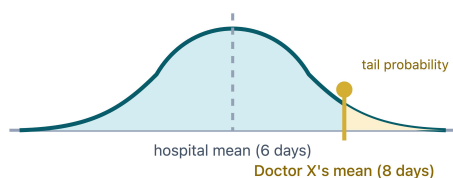
5. The sampling distribution of the sample mean has its own spread. What do we call the standard deviation of that sampling distribution?

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n - 1}}$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

- a) Population standard deviation
- b) Standardization Exponent
- c) Standard Error ✓

6. Doctor X's 25 recent patients had a mean recovery time of 8 days, while the hospital's mean is 6 days. In inferential statistics, to understand if this is a significant difference, we ask:



- a) How often would random samples of 25 produce a mean this high or higher just by chance? ✓
- b) What is the median expectation of Doctor X's patients compared to the median of the population